Revisiting the Mechanical Advantage of Compliant Mechanisms

iNaCoMM 2025

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December 08, 2025

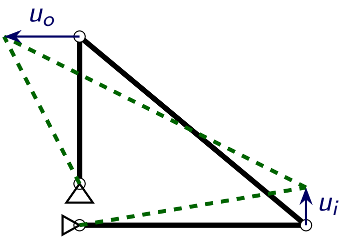
Outline

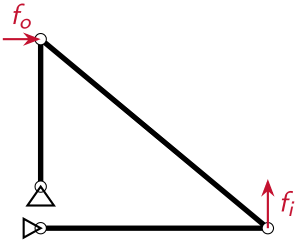
Introduction

For a single-input single-output mechanisms,

- **Displacement advantage** (DA) refers to the ratio of the displacement of the designated output and input degrees of freedom,
- **Mechanical advantage** (MA) refers to the ratio of force experienced along the designated output and input degrees of freedom.

$$\mathrm{DA} = \frac{u_o}{u_i} \qquad \mathrm{MA} = \frac{f_o}{f_i}$$





Revisiting MA in compliant mechanisms

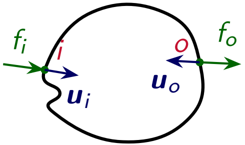
In Salamon and Midha (1998)*:

Initial state:

$$f_o = f_o^\dagger$$
$$f_i = f_i^\dagger$$
$$u_i = u_i^\dagger$$
$$u_o = u_o^\dagger$$
$$SE = SE^\dagger$$

Perturbed state:

$$f_o = f_o^\dagger + \Delta u_i^\dagger$$
$$f_i = f_i^\dagger$$
$$u_i = u_i^\dagger + \Delta u_i^\dagger$$
$$u_o = u_o^\dagger + \Delta u_o^\dagger$$
$$SE = SE^\dagger + \Delta SE^\dagger$$
$$\Delta SE^\dagger = f_i^\dagger \Delta u_i^\dagger - f_o^\dagger \Delta u_o^\dagger$$



$$\boxed{\mathrm{MA} = \frac{f_0^\dagger}{f_i^\dagger} = \frac{\Delta u_i^\dagger}{\Delta u_o^\dagger} - \frac{\Delta SE^\dagger}{f_i^\dagger \Delta u_o^\dagger} \equiv \mathrm{MA}_r - \mathrm{MA}_c}$$

*B. A. Salamon and A. Midha. "An Introduction to Mechanical Advantage in Compliant Mechanisms". In: *Journal of Mechanical Design* 120.2 (June 1998), pp. 311–315.

Revisiting MA in compliant mechanisms

$$\boxed{\mathrm{MA} = \frac{\Delta u_i^\dagger}{\Delta u_o^\dagger} - \frac{\Delta U^\dagger}{f_i^\dagger \Delta u_o^\dagger} \equiv \mathrm{MA}_r - \mathrm{MA}_c}$$

Excerpt from Salamon and Midha (1998):
"The first term in Eq. (14) takes the form of a **rigid-body mechanical advantage***. This term would result if an* **instant center analysis** *for mechanical advantage (Shigley and Uicker, 1980) could be applied to the compliant mechanism in any instantaneous position..."*

Revisiting MA in compliant mechanisms

$$\boxed{\mathrm{MA} = \frac{\Delta u_i^{\dagger}}{\Delta u_o^{\dagger}} - \frac{\Delta SE^{\dagger}}{f_i^{\dagger}\Delta u_o^{\dagger}} \equiv \mathrm{MA}_r - \mathrm{MA}_c}$$

"The second term in Eq. (14) also resembles a mechanical advantage term. It is referred to as the **compliant component of the mechanical advantage** *(MA_c), and it accounts for the energy stored in the mechanism..."*

Outline

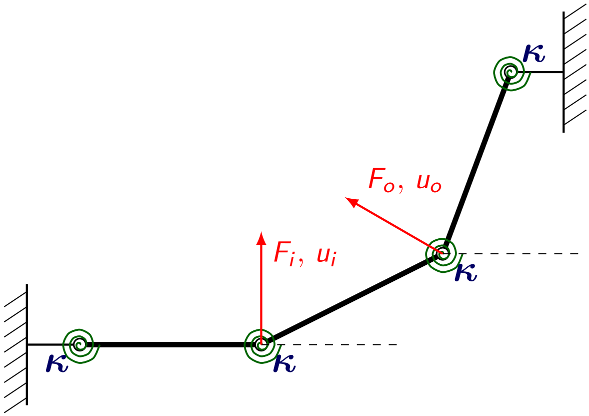
1. Introduction

2. **Mechanical advantage based on kinetoelastics**

3. Comparing rigid-body and compliant mechanisms

4. On Prof. Midha's famous compliant crimper

Kineoelastics



On κ

$\kappa \to 0 \implies$ 4-bar rigid-body linkage

$\kappa \to \infty \implies$ compliant 4-bar linkage

MA based on kinetoelastics

$$[\boldsymbol{K_t}(\boldsymbol{u})]\{\boldsymbol{\Delta u}\} = \{\boldsymbol{\Delta f}\} \iff \{\boldsymbol{\Delta u}\} = [\boldsymbol{C_t}(\boldsymbol{u})]\{\boldsymbol{\Delta f}\}$$

For a single-input single-output mechanism:

$$\Delta u_i = [C_t(\boldsymbol{u})]_{ii}\Delta f_i - [C_t(\boldsymbol{u})]_{io}\Delta f_o$$
$$\Delta u_o = [C_t(\boldsymbol{u})]_{oi}\Delta f_i - [C_t(\boldsymbol{u})]_{oo}\Delta f_o$$

$$\Delta f_o = \frac{[C_t(\boldsymbol{u})]_{oi}\Delta f_i}{[C_t(\boldsymbol{u})]_{oo}} - \frac{\Delta u_o}{[C_t(\boldsymbol{u})]_{oo}}$$

Integrating the last expression from **initial state** $(1 \equiv \{u_i = 0, u_o = 0, f_i = 0, f_o = 0\})$ to its **final state** $(2 \equiv \{u_i = u_i^\dagger, u_o = u_o^\dagger, f_i = f_i^\dagger, f_o = f_o^\dagger\})$:

$$\boxed{f_o^\dagger = \int_1^2 \frac{[C_t(\boldsymbol{u})]_{oi} df_i}{[C_t(\boldsymbol{u})]_{oo}} - \int_1^2 \frac{du_o}{[C_t(\boldsymbol{u})]_{oo}}}$$

MA based on kinetoelastics

The last expression can be rewritten in the following form:

$$\frac{f_o^\dagger}{f_i^\dagger} = \mathrm{MA}_s \left(1 - \frac{f_n}{f_i^\dagger}\right)$$

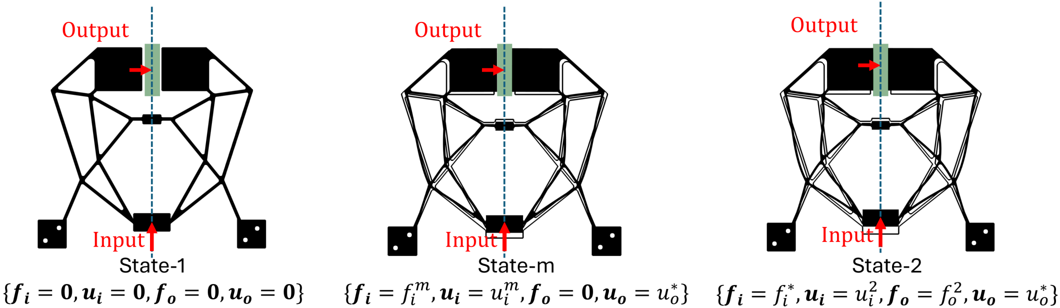
$$f_n \equiv \left(\frac{\int_1^2 \frac{du_o}{[C_t(\boldsymbol{u})]_{oo}}}{\frac{1}{f_i^\dagger} \int_1^2 \frac{[C_t(\boldsymbol{u})]_{oi} df_i}{[C_t(\boldsymbol{u})]_{oo}}} \right), \qquad \mathrm{MA}_s = \frac{1}{f_i^\dagger} \int_1^2 \frac{[C_t(\boldsymbol{u})]_{oi} \; df_i}{[C_t(\boldsymbol{u})]_{oo}}$$

MA based on kinetoelastics

Let us define the intermediate stage where the mechanism begins to engage the workpiece as **state-m** $\equiv \{u_i^m, u_o^\dagger, f_i^m, f_o = 0\}$.

$$\frac{f_o^\dagger}{f_i^\dagger} = \frac{1}{f_i^\dagger}\int_1^2 \frac{[C_t(\boldsymbol{u})]_{oi}\ df_i}{[C_t(\boldsymbol{u})]_{oo}} - \frac{1}{f_i^\dagger}\int_1^m \frac{[C_t(\boldsymbol{u})]_{oi}\ df_i}{[C_t(\boldsymbol{u})]_{oo}} = \frac{1}{f_i^\dagger}\int_{f_i^m}^{f_i^\dagger} \frac{[C_t(\boldsymbol{u})]_{oi}\ dF_i}{[C_t(\boldsymbol{u})]_{oo}},$$

where f_i^m is the input force at which the mechanism engages the workpiece.



MA based on kinetoelastics

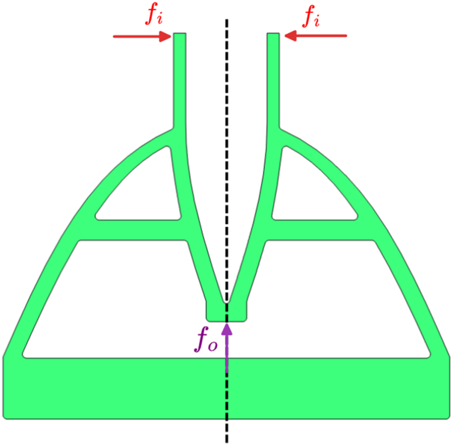
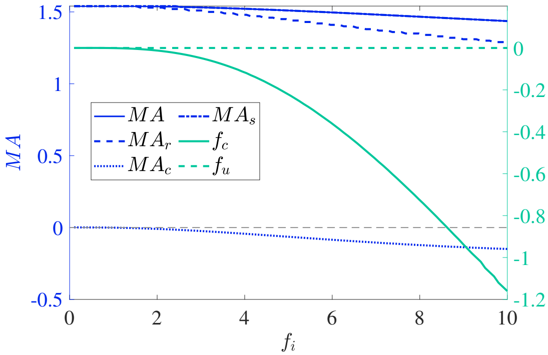
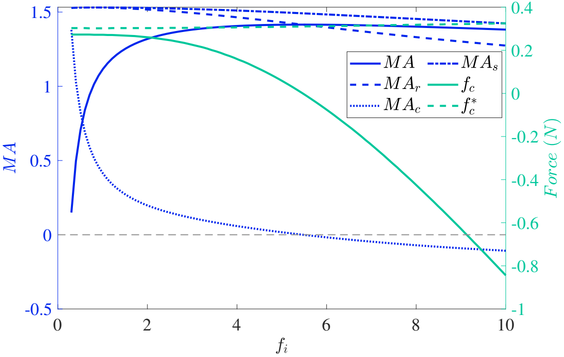


Figure: A compliant crimping mechanism

MA based on kinetoelastics



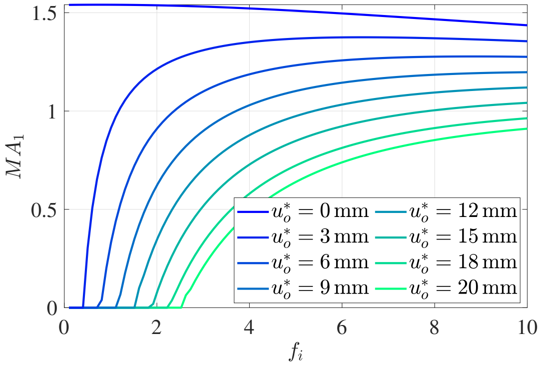
(a) Variability of the different MA terms and compliance forces with input force for $u_o^\dagger = 0$



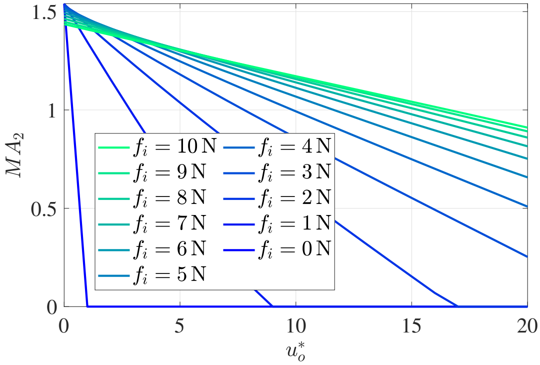
(b) Variability of the different MA terms and compliance forces with input force for $u_o^\dagger = 2$ mm

MA based on kinetoelastics

$$\frac{f_o^\dagger}{f_i^\dagger} = \frac{1}{f_i^\dagger}\int_1^2 \frac{[C_t(\boldsymbol{u})]_{oi}\ df_i}{[C_t(\boldsymbol{u})]_{oo}} - \frac{1}{f_i^\dagger}\int_1^2 \frac{du_o}{[C_t(\boldsymbol{u})]_{oo}}$$



(a) MA_1



(b) MA_2

MA based on kinetoelastics

Main takeaways

1. It can be summarised that the MA in general is **maximum when the mechanism behaves like a structure** (output port is fixed),
2. There is a **loss in MA** due to the deformation the system must experience to **engage the workpiece**.
3. The **MA of the structure** at the particular load is the **average of** MA_r (the ratio of incremental displacement of input and output) **over the entire motion**.

Eigenvalue-based decomposition of DA and MA

$$[\boldsymbol{K}]\{\boldsymbol{u}\} = \{\boldsymbol{f}\}$$

eigenvalues: $(\lambda_k \forall k \in \{1, 2..n\})$
orthonormal eigenvectors: $(\boldsymbol{a_k} \forall k \in \{1, 2..n\})$

$$\{\boldsymbol{u}\} = \Sigma_{k=1}^{n} \frac{(\boldsymbol{a_k} \otimes \boldsymbol{a_k})\{\boldsymbol{f}\}}{\lambda_k}$$

Where, $\otimes$ denotes dyadic product, defined as:

$$\boldsymbol{b} \otimes \boldsymbol{c} \equiv (b \otimes c)_{ij} = b_i \cdot c_j$$

Eigenvalue based decomposition of DA and MA

For single-input single-output mechanisms:

$$u_i = \Sigma_{k=1}^{n} \frac{f_i (a_k \otimes a_k)_{ii}}{\lambda_k} - \frac{f_o (a_k \otimes a_k)_{io}}{\lambda_k}$$
$$u_o = \Sigma_{k=1}^{n} \frac{f_i (a_k \otimes a_k)_{io}}{\lambda_k} - \frac{f_o (a_k \otimes a_k)_{oo}}{\lambda_k}$$

The last relation can be simplified to:

$$\frac{f_o}{f_i} = \mathrm{MA}_s \left(1 - \frac{f_n}{f_i}\right)$$

$$f_n \equiv \frac{u_o}{\Sigma_{k=1}^{n} \frac{(a_k \otimes a_k)_{io}}{\lambda_k}}, \qquad \mathrm{MA}_s = \frac{\Sigma_{k=1}^{n} \frac{(a_k \otimes a_k)_{io}}{\lambda_k}}{\Sigma_{k=1}^{n} \frac{(a_k \otimes a_k)_{oo}}{\lambda_k}}$$

Eigenvalue based decomposition of DA and MA

Similarly, it is possible to represent the displacement advantage as:

$$\frac{u_o}{u_i} = \frac{\sum_{k=1}^{n} \frac{(a_k \otimes a_k)_{io}}{\lambda_k}}{\sum_{k=1}^{n} \frac{(a_k \otimes a_k)_{ii}}{\lambda_k}} \left(1 - \frac{u_c}{u_i}\right)$$

$$u_c \equiv \frac{f_o \left(\left(\sum_{k=1}^{n} \frac{(a_k \otimes a_k)_{ii}}{\lambda_k}\right) \left(\sum_{k=1}^{n} \frac{(a_k \otimes a_k)_{oo}}{\lambda_k}\right) - \left(\sum_{k=1}^{n} \frac{(a_k \otimes a_k)_{io}}{\lambda_k}\right)^2 \right)}{\left(\sum_{k=1}^{n} \frac{(a_k \otimes a_k)_{io}}{\lambda_k}\right)}$$

Eigenvalue based decomposition of DA and MA

DA_f can be rewritten as:

$$\mathrm{DA}_f = \frac{\sum_{k=1}^{n} \frac{(a_k \otimes a_k)_{io}}{\lambda_k}}{\sum_{k=1}^{n} \frac{(a_k \otimes a_k)_{ii}}{\lambda_k}} = \frac{\sum_{k=1}^{n} \frac{(a_k)_i \times (a_k)_o}{\lambda_k}}{\sum_{k=1}^{n} \frac{(a_k)_i \times (a_k)_i}{\lambda_k}} = \frac{\sum_{k=1}^{n} \frac{(a_k)_i^2}{\lambda_k} \mathrm{DA}_k}{\sum_{k=1}^{n} \frac{(a_k)_i^2}{\lambda_k}},$$

where DA_k is the ratio $(a_k)_o/(a_k)_i$ for k^{th} eigenvector. Also,

$$\mathrm{MA}_s = \frac{\sum_{k=1}^{n} \frac{(a_k \otimes a_k)_{io}}{\lambda_k}}{\sum_{k=1}^{n} \frac{(a_k \otimes a_k)_{oo}}{\lambda_k}} = \frac{\sum_{k=1}^{n} \frac{(a_k)_i \times (a_k)_o}{\lambda_k}}{\sum_{k=1}^{n} \frac{(a_k)_o \times (a_k)_o}{\lambda_k}} = \frac{\sum_{k=1}^{n} \frac{(a_k)_o^2}{\lambda_k} \left(\frac{1}{\mathrm{DA}_k}\right)}{\sum_{k=1}^{n} \frac{(a_k)_o^2}{\lambda_k}}$$

Eigenvalue based decomposition of DA and MA

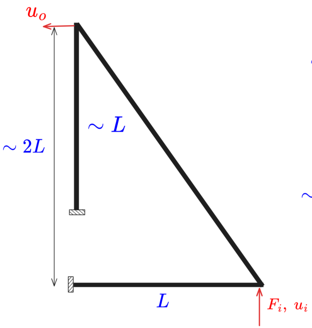
Consequently,

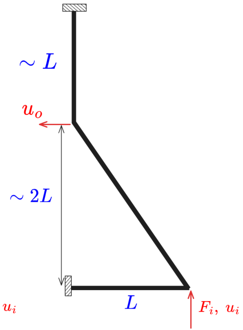
$$\mathrm{DA} \times \mathrm{MA} \leq \mathrm{MA}_s \times \mathrm{DA}_f = \frac{\left(\sum_{k=1}^{n} \frac{(a_k)_i^2}{\lambda_k} \mathrm{DA}_k\right)^2}{\left(\sum_{k=1}^{n} \frac{(a_k)_i^2}{\lambda_k} \mathrm{DA}_k^2\right)\left(\sum_{k=1}^{n} \frac{(a_k)_i^2}{\lambda_k}\right)} \leq 1$$

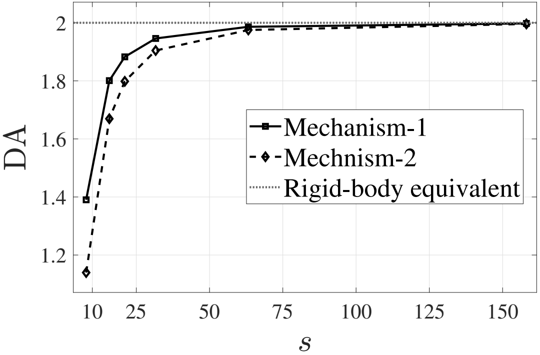
$\mathrm{MA}_s \times \mathrm{DA}_f = 1$

The equality is only satisfied when $\lambda_1 = 0$!

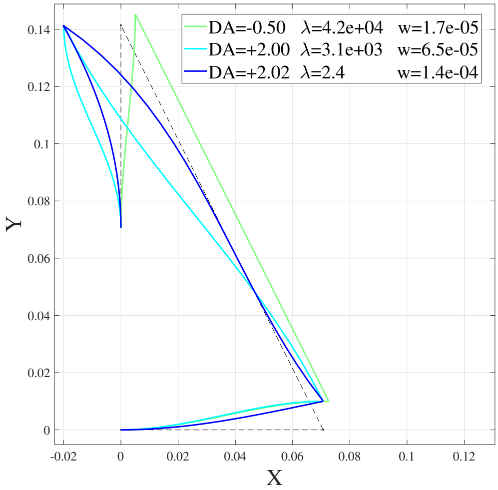
Eigenvalue based decomposition of DA and MA



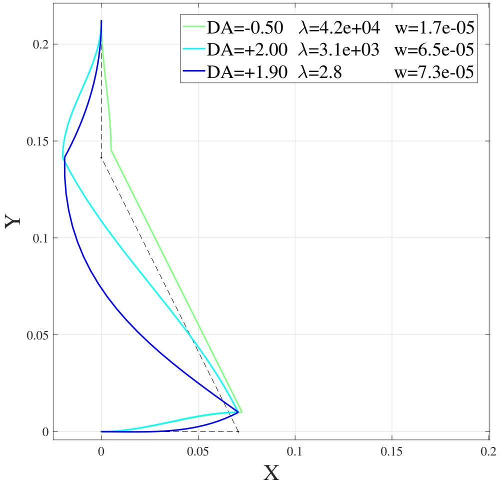




Eigenvalue based decomposition of DA and MA



(a) Salient eigenvectors of mechanism-1 ($s = 15.8$, $\mathrm{DA}_f = 1.80$)



(b) Salient eigenvectors of mechanism-2 ($s = 15.8$, $\mathrm{DA}_f = 1.67$)

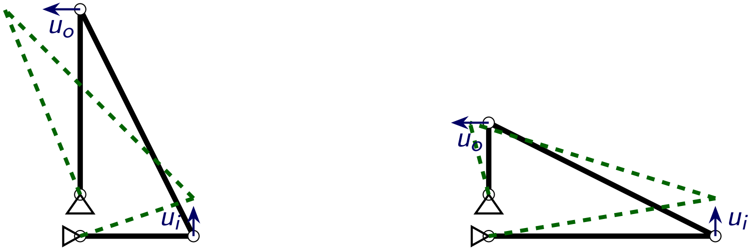
Outline

Comparing rigid-body and compliant mechanisms

In rigid-body mechanisms, we know

$$\mathrm{MA} = 1/\mathrm{DA}$$

We also know that DA and MA get swapped when we flip output and input:



(a) DA=2, MA=0.5 (b) DA=0.5, MA=2

Comparing rigid-body and compliant mechanisms

Now we know,

$$\mathrm{DA} \times \mathrm{MA} \leq \mathrm{DA}_f \times \mathrm{MA}_s < 1$$

However, DA_f can be rewritten as:

$$\mathrm{DA}_f = \frac{\sum_{k=1}^{n} \frac{(a_k)_i \times (a_k)_o}{\lambda_k}}{\sum_{k=1}^{n} \frac{(a_k)_i \times (a_k)_i}{\lambda_k}} \qquad \mathrm{MA}_s = \frac{\sum_{k=1}^{n} \frac{(a_k)_i \times (a_k)_o}{\lambda_k}}{\sum_{k=1}^{n} \frac{(a_k)_o \times (a_k)_o}{\lambda_k}}$$

Still, the DA_f and MA_s get swapped when input and output are flipped.

Comparing rigid-body and compliant mechanisms

Uniqueness of MA and DA

- MA or DA can't be stated uniquely for a compliant mechanism!
  ⋆To define MA and DA uniquely (for small deformations) for a given compliant mechanism, we must specify at least **two of the following four parameters**: $\{f_i, f_o, u_i, u_o\}$.
- This is also true for a rigid-body mechanism $(\lambda_1 = 0)$,
  ⋆MA: u_i and u_o being zero (a consequence of **static equilibrium**),
  ⋆DA: f_i and f_o being zero (**mobility and degree of freedom**).

Combining compliant mechanisms

Imagine two compliant mechanisms with spring lever parameters $k_i^1,\ k_o^1,\ n_1$ and $k_i^2,\ k_o^2,\ n_2$.

$$\begin{bmatrix} k_i^1 + (n_1)^2 k_o^1 & -n_1 k_o^1 & 0 \\ -n_1 k_o^1 & +k_i^2 + (n_2)^2 k_o^2 + k_o^1 & -n_2 k_o^2 \\ 0 & -n_2 k_o^2 & k_o^2 \end{bmatrix} \begin{Bmatrix} u_i^1 \\ u_o^1 \text{ or } u_i^2 \\ u_o^2 \end{Bmatrix} = \begin{Bmatrix} f_i^1 \\ f_o^1 \text{ or } f_i^2 \\ f_o^2 \end{Bmatrix}$$

Now, eliminating the intermediate degree of freedom (no load at this point):

$$\frac{1}{k_o^2 n_2^2 + k_i^2 + k_o^1} \begin{bmatrix} k_o^1 k_o^2 n_1^2 n_2^2 + k_i^2 k_o^1 n_1^2 + k_i^2 k_o^1 n_2^2 + k_i^1 k_i^2 + k_i^1 k_o^1 & -k_o^1 k_o^2 n_1 n_2 \\ -k_o^1 k_o^2 n_1 n_2 & k_o^2 (k_i^2 + k_o^1) \end{bmatrix} \begin{Bmatrix} u_i^1 \\ u_o^2 \end{Bmatrix} = \begin{Bmatrix} f_i^1 \\ f_o^2 \end{Bmatrix}$$

The equivalent DA for this mechanism is:

$$n^\dagger = \left(\frac{k_o^1}{k_i^2 + k_o^1} \right) n_1 n_2$$

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Eigenvalue based decomposition of DA and MA

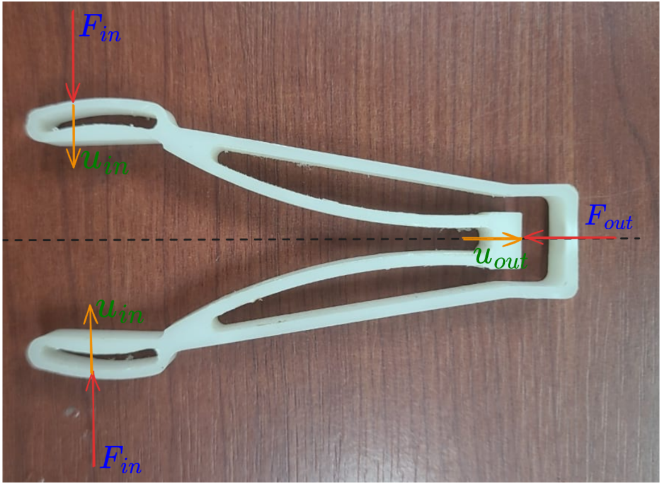
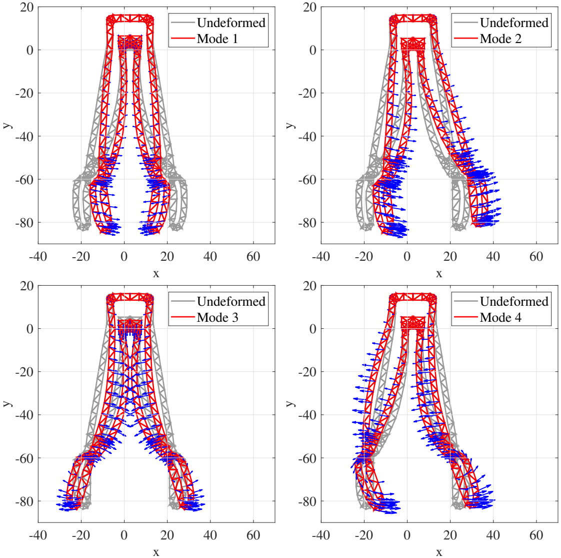


Figure: The first compliant mechanism designed by Prof. Ashok Midha

$$\mathrm{MA}_s = 5.6,$$
$$\mathrm{DA}_f = 0.10,$$
$$\mathrm{MA}_s \times \mathrm{DA}_f = 0.56$$

Eigenvalue-based decomposition of DA and MA



$$w_1 \times \mathrm{MA}_1 = 5.42$$
$$w_2 \times \mathrm{MA}_2 = 0.06$$
$$w_3 \times \mathrm{MA}_3 = 0.14$$
$$w_4 \times \mathrm{MA}_4 = -0.01$$
$$\vdots$$
$$Total = 5.58$$

Thank you!